Problem Statement : 3

AI-Powered Separation of Overlapping Stamps and Signatures

Team-Name : Minecrafters

Team Members

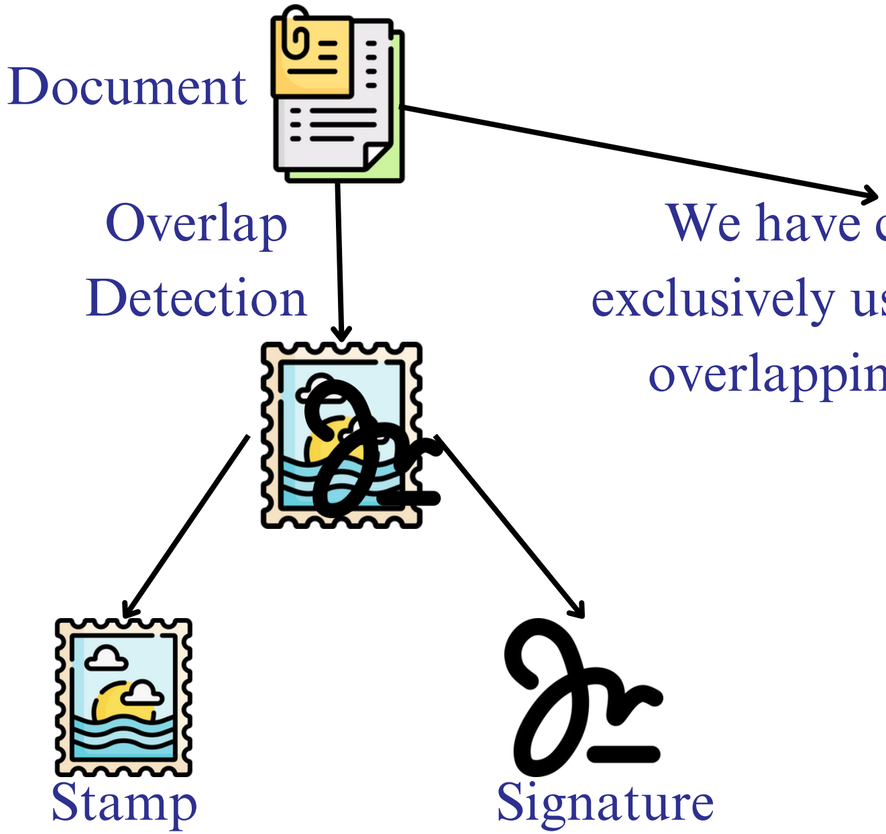
- Varad Dherange
- Yasir Ahmad
- Kumar Shivam
- Arin Yadav

Solution Approaches :

- Using Visual Transformers
  - **Meta :** SAM-2
  - **Microsoft :** Florence-2
- Semantic Segmentation
- Keypoint Detection & Feature Extraction

Idea Title

Signature & Stamp Semantic Segmentation



Step 1 : Overlapping Signature & Stamp

Document



(Region of Interest Detection)



Overlapping Stamp & Signature in Document

(Done as Bounding Box)



Example from the dataset

YOLO v12 : Computer Vision model for Stamp & Signature Detection

Microsoft/Florence-2-base : Finetuned vision transformer on dataset for Object detection

Approach for the current step in overlap detection.

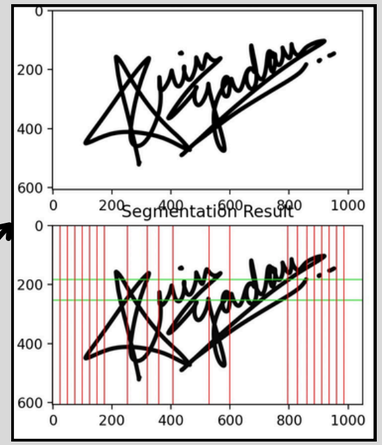
Step 2 : Data Preparation and Image Processing

(Bounding Box)



Example from the dataset

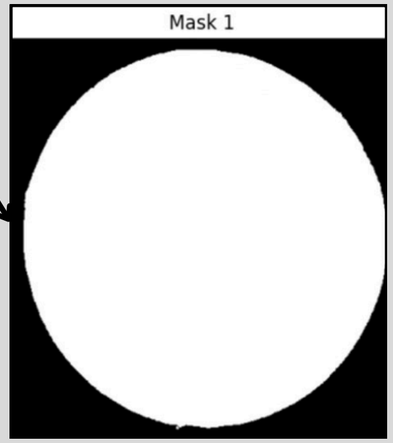
Obtaining the bounding box, we can proceed with the precise segmentation of the signature



Evaluating

- **Feature extraction**
- **keypoint detection**

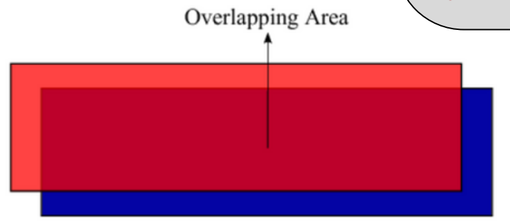
techniques for isolating the signature from overlapping entities.



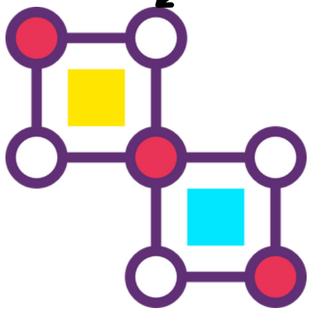
Evaluating

- **Masking and**
- **Classification capabilities**

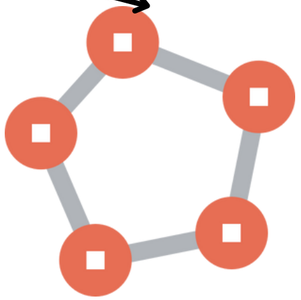
using Vision Transformer models for precise data segmentation.



The attempted approach lacked precision in segmenting the overlapping entities.

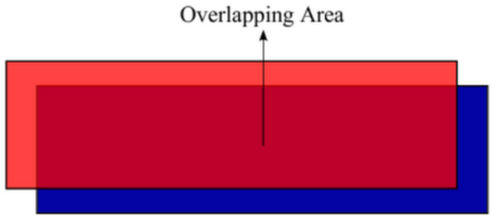


Bounding box annotation for enhanced overlap detection.



Polygon annotation for precise signature extraction.

Step 3: Semantic Segmentation for Precise Differentiation

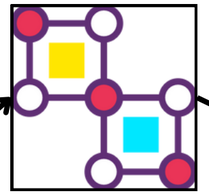


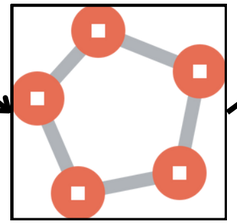


Original image after detecting overlapping entities.

Annotated Actual data







Annotated data for both signature segmentation and stamp segmentation

SegFormer

NVIDIA

Simple and Efficient Design for Semantic Segmentation with Transformers

Utilizing the prepared dataset to fine-tune the **NVIDIA Transformer** model for **semantic segmentation**.

Predicted Mask Overlay from Model



The region containing the signature is masked and segmented for precise extraction

Step 4: Semantic Segmentation Output

Actual Mask Overlay



SegFormer

NVIDIA

Simple and Efficient Design for Semantic Segmentation with Transformers

Predicted Mask Overlay





Input Overlap Image

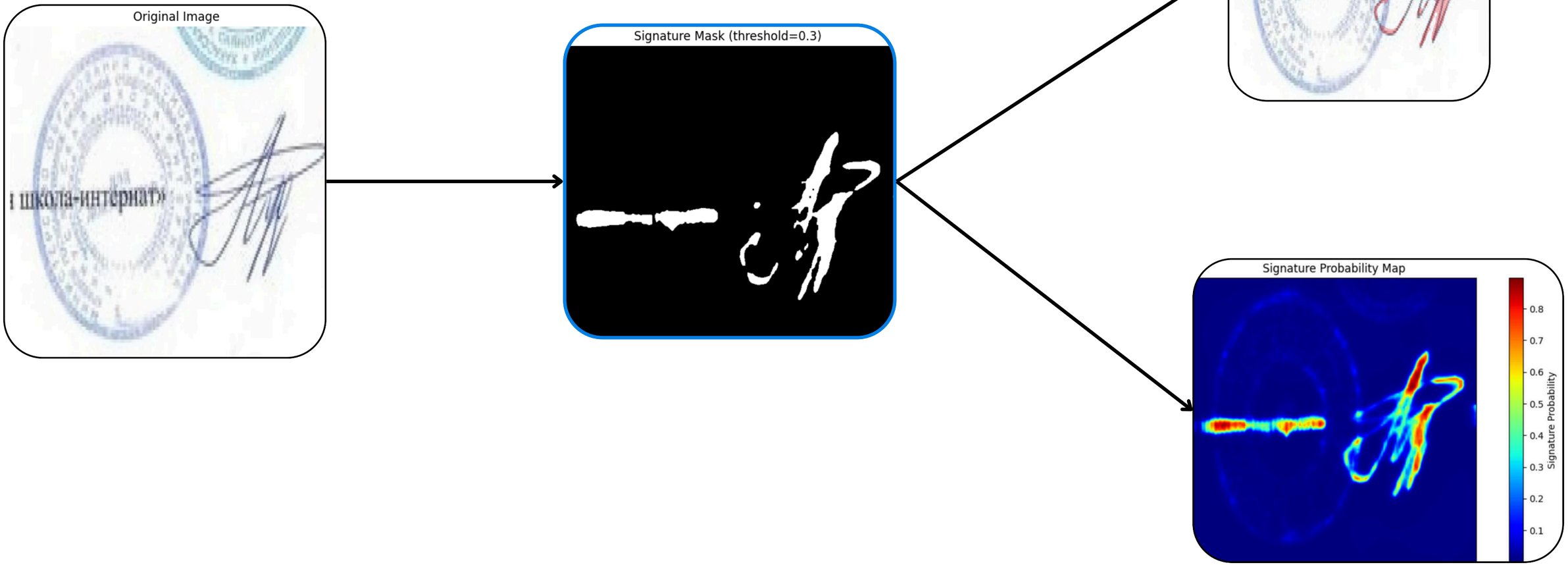


Final Extracted Signature Segment

Final Extracted Stamp Segment



Input Image → Processed Stamp → Processed Stamp minus Segmentation Mask = Final Output


Original Image
Signature Mask (threshold=0.3)
Signature Overlay
Signature Probability Map
Signature Probability
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1

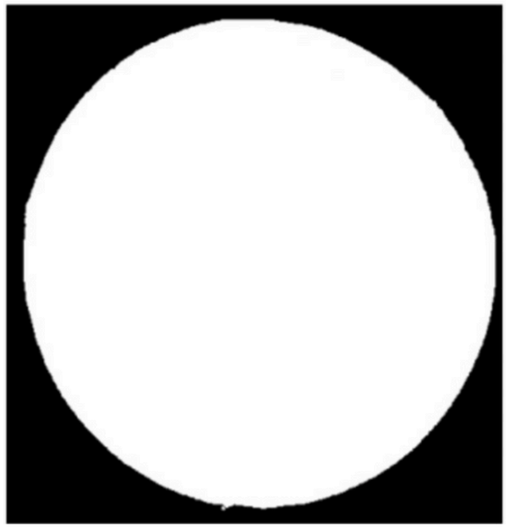
Preliminary Methodologies



Stamp Detection

Background Removal

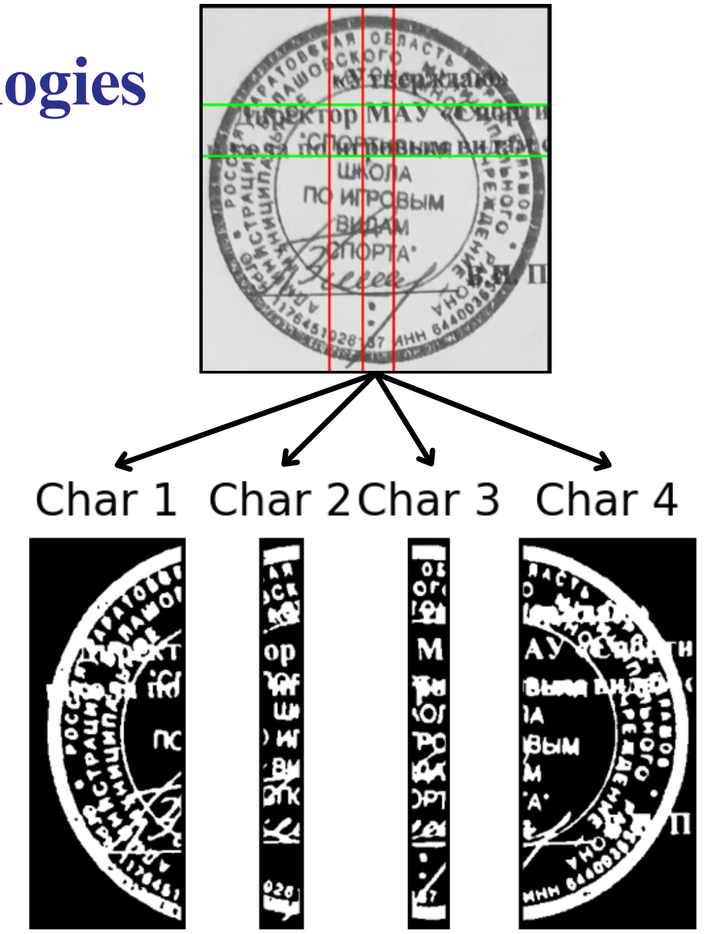
This lacked the ability to detect and segment Stamps



Masking

Region of Overlapped Entity

Due to how Masking works it was only able to segment off the Outer Circle as one entity



Cursive Overlapped Character Segmentation

Using Otsu's algorithm, core-zone detection for improved segmentation of slanted and overlapping words & identifies character boundaries.

Too much noise around the Stamp causes this to not work as intended.

Masking

Finegrain Enterprise Company
https://finegrain.ai finegrain_ai finegrain-ai

AI at Meta Enterprise Company Verified
https://ai.facebook.com/ facebookresearch

Segmentation

BRIA AI Enterprise Company
https://bria.ai/ bria_ai_ Bria-AI

ZhengPeng7/BiRefNet

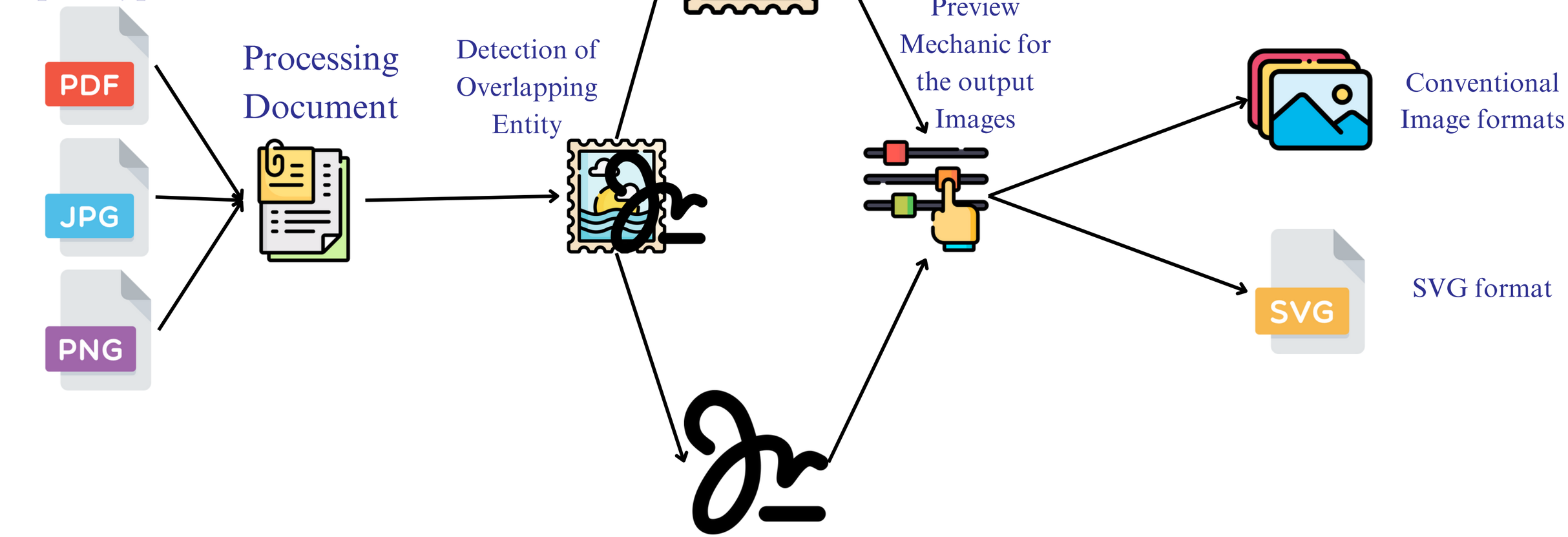
Keypoint Extraction & Feature Extraction

University of Sydney Community
https://www.sydney.edu.au/

facebook/dinov2-base

Open Source Models

The following models were finetuned to do the specific tasks but without proper data they could not perform the operations


Input Types
PDF
JPG
PNG
Processing Document
Detection of Overlapping Entity
Preview Mechanic for the output Images
Output Types
Conventional Image formats
SVG
SVG format

THANK YOU